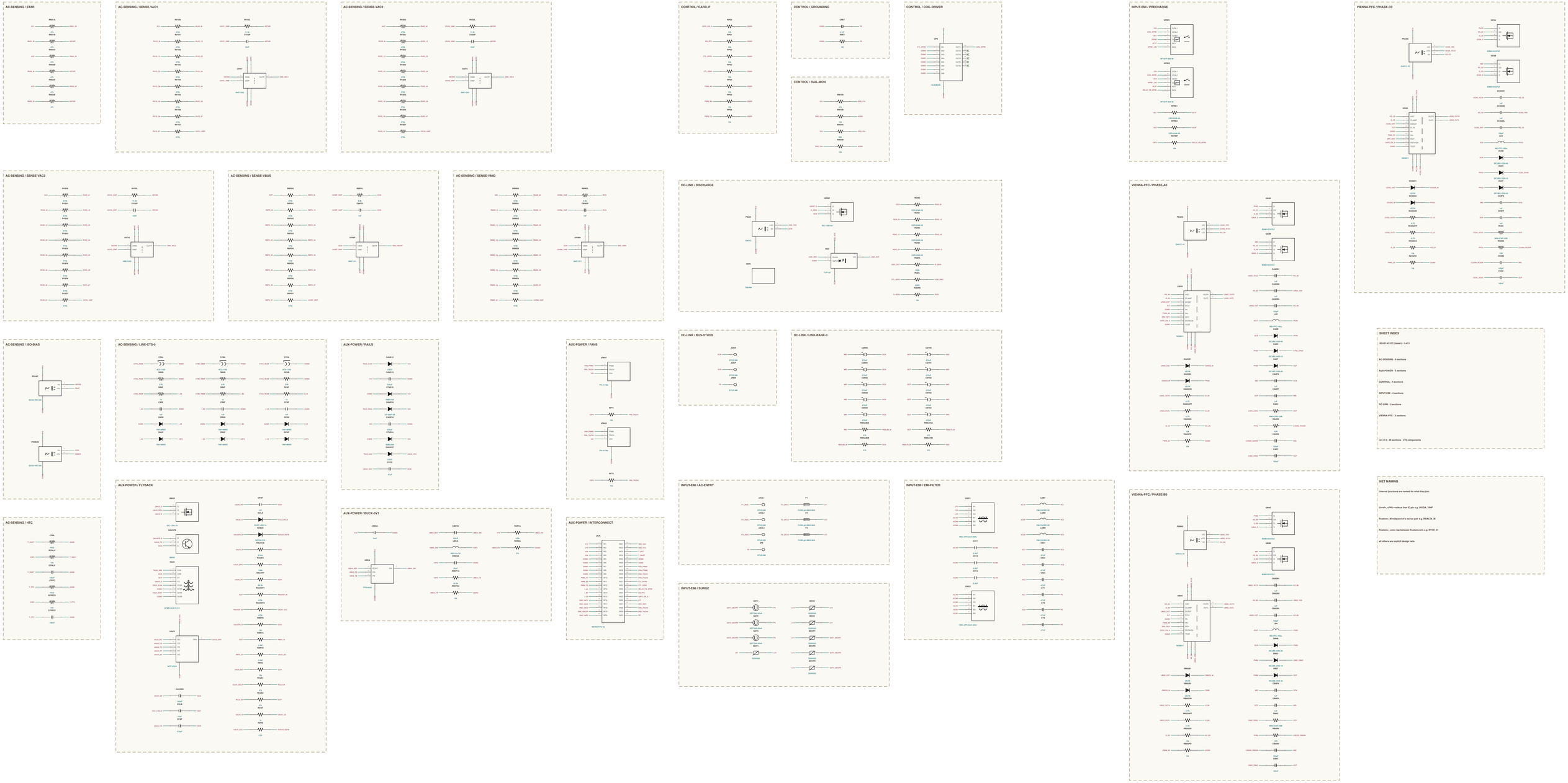
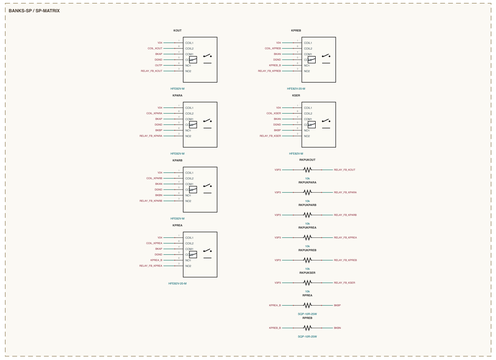


DC-Modules 30 kW — AC-DC board (Vienna PFC)

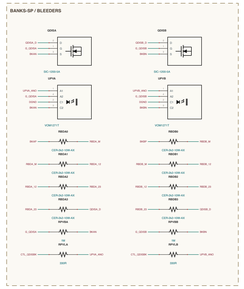
DC-Modules 30 kW - board AC-DC (lower), sheet 1 of 3

2026-09-06 · rev D.3

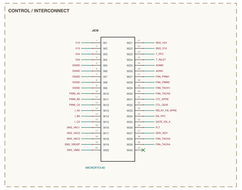
BANKS-SP / SP-MATRIX



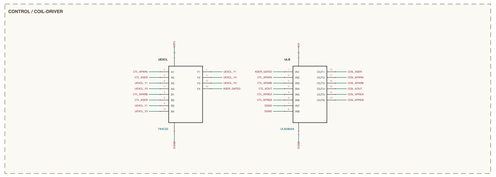
BANKS-SP / BLEEDERS



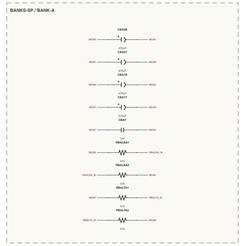
CONTROL / INTERCONNECT



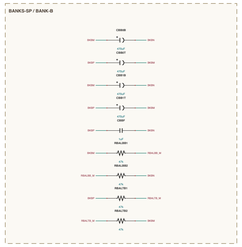
CONTROL / COL-DRIVER



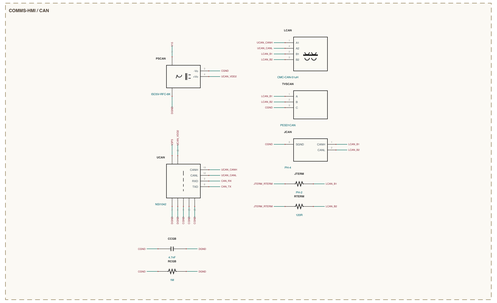
BANKS-SP / BANK-A



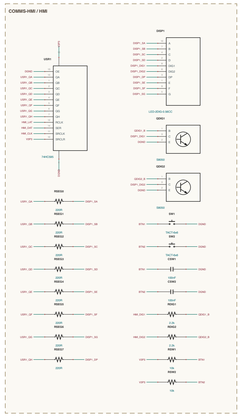
BANKS-SP / BANK-B



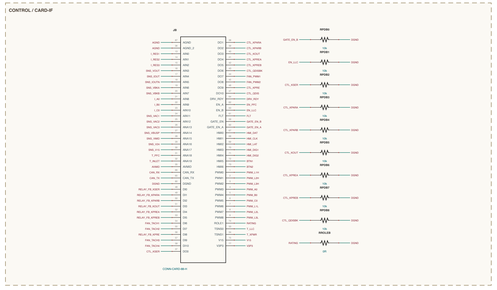
COMMS-HMI / CAN



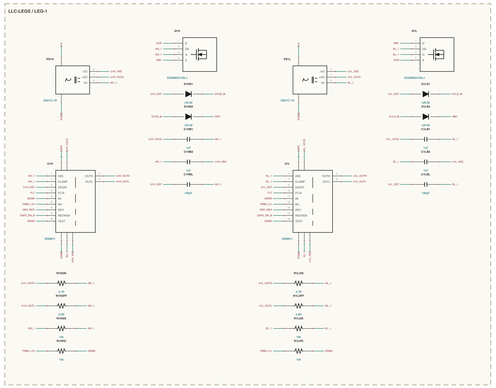
COMMS-HMI / HMI



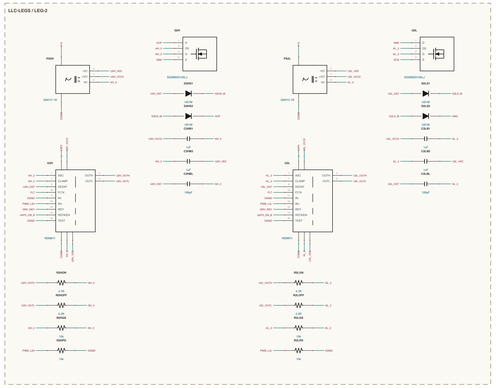
CONTROL / CARD-IF



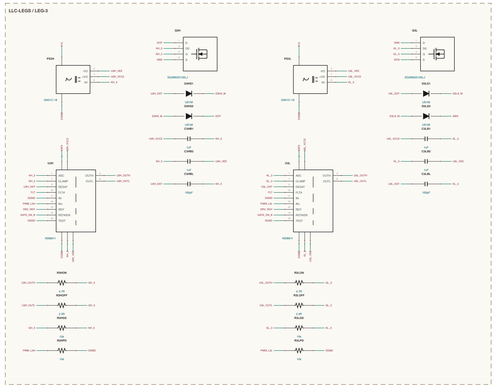
LLC-LEGS / LEG-1



LLC-LEGS / LEG-2

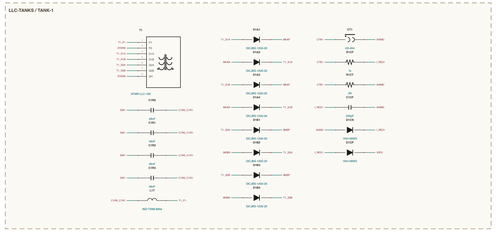


LLC-LEGS / LEG-3

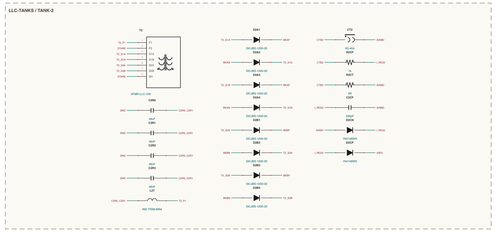


LLC-LEGS / INR-IN

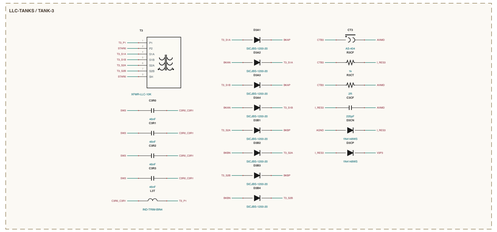
LLC-TANKS / TANK-1



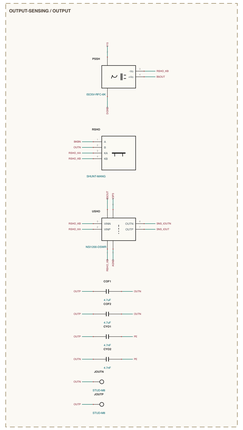
LLC-TANKS / TANK-2



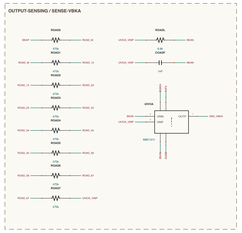
LLC-TANKS / TANK-3



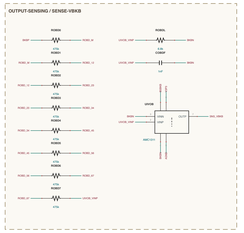
OUTPUT-SENSING / OUTPUT



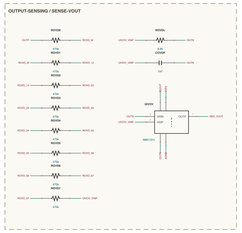
OUTPUT-SENSING / SENSE-VBKA



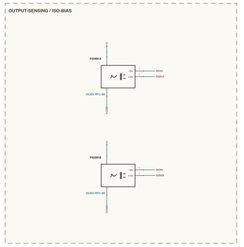
OUTPUT-SENSING / SENSE-VBKB



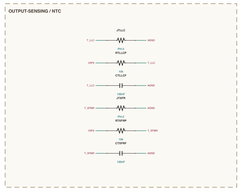
OUTPUT-SENSING / SENSE-VOUT



OUTPUT-SENSING / ISO-BIAS



OUTPUT-SENSING / NTC



SHEET INDEX

30 kW DC-DC (upper) - 3 of 3

OUTPUT-SENSING - 6 sections

BANKS-SP - 4 sections

LLC-LEGS - 4 sections

CONTROL - 3 sections

LLC-TANKS - 3 sections

COMMS-HMI - 2 sections

rev D.3 - 32 sections - 279 components

NET NAMING

internal junctions are named for what they join

(xxxh)_uPfbs node at that IC pin e.g. UV12A_VINP

Rxxbxxx_W resistpoint of a series pair e.g. R84a7A_W

Rxxbxxx_xxxx-top between Rxxbxxxxh-x.g. R5Y2E_D1

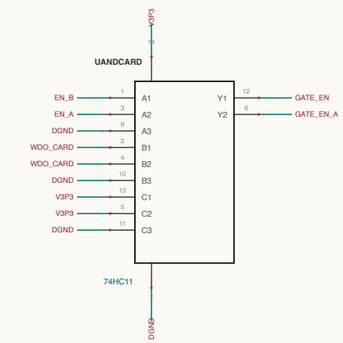
all others are explicit design nets

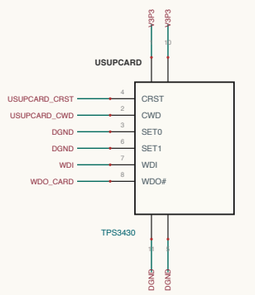
DC-Modules 30 kW — DC-DC board (3-phase LLC)

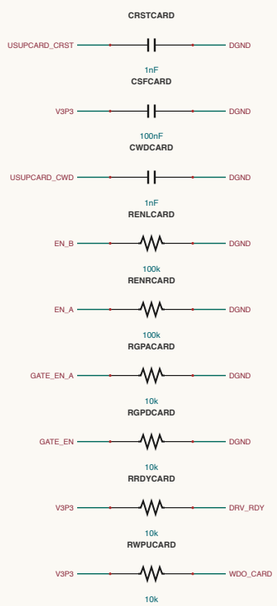
DC-Modules 30 kW - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.3

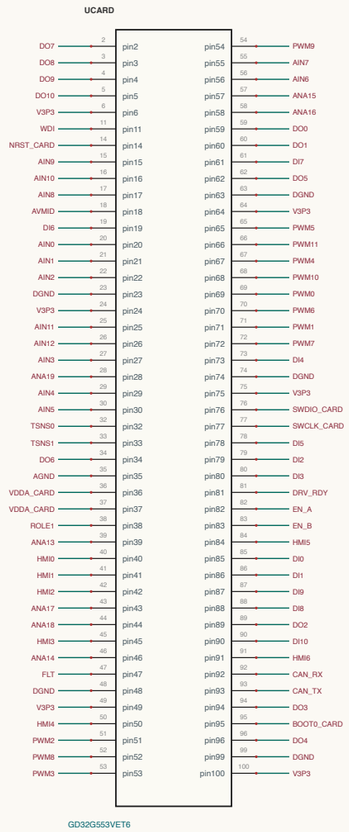
CONTROL / SAFETY

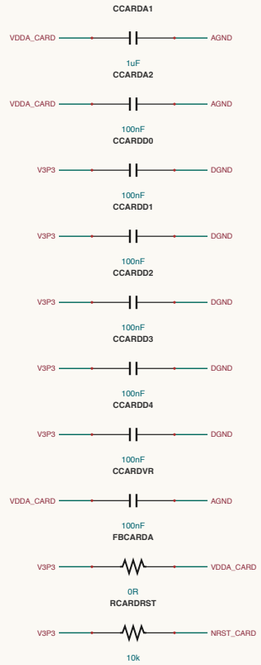




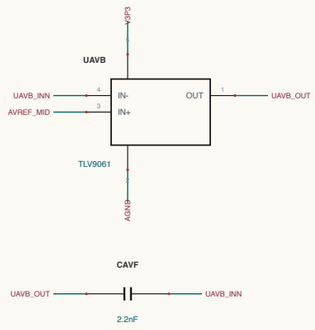


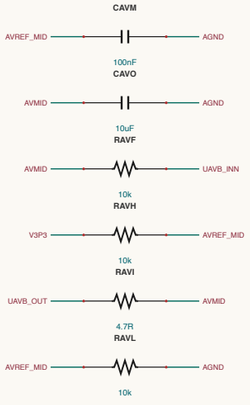
CONTROL / MCU



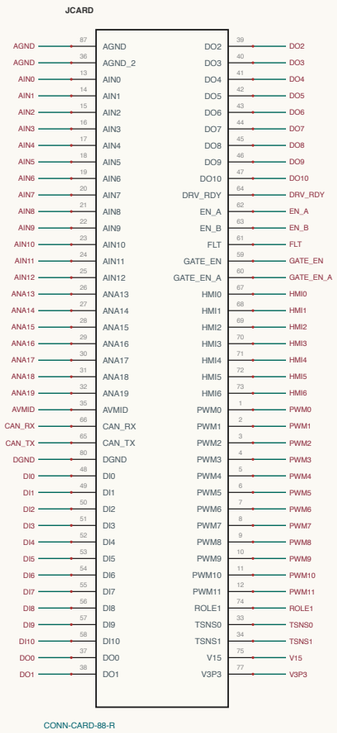


CONTROL / ANALOG-MID

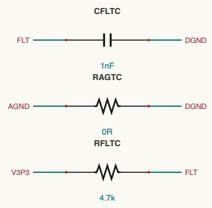




CONTROL / CARD-IF

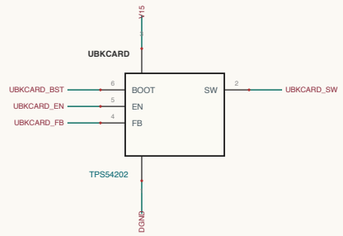


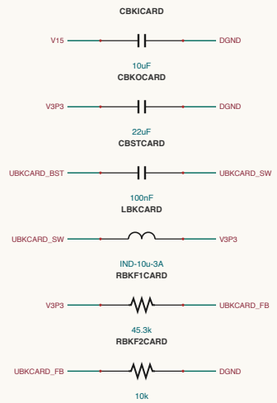
CONTROL / FLT-GROUND



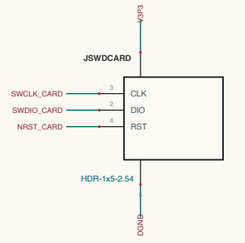
CONTROL / ROLE

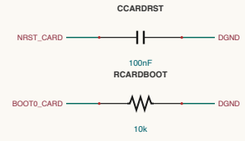
CONTROL / BUCK-3V3





CONTROL / SWD-BOOT





30/60 kW Control card - 3 of 3 · rev D.3 · 8 sections · 45 components

NET NAMING: U<ref>_<PIN> = node at that IC pin · R<stem>_M = series-pair midpoint · R<stem>_<nm> = tap between R<stem>n/m · all others are explicit design nets

DC-Modules — Control Card (GD32G553VET6, one card for both converter roles)

DC-Modules 30/60 kW - board Control card, sheet 3 of 3

2026-09-06 · rev D.3